



الجامعة الإسلامية - غزة
كلية تكنولوجيا المعلومات
ويب 2 عملي
SDEV 2106, WDM 2010,
MOBC 2102
الفصل الثاني: 2025/2026
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PHP Basics — Advanced Assignment

10 Mixed-Topic Challenges + AI Integration Guide

Part 1 — How to Use AI Responsibly:

Artificial intelligence is a powerful learning tool — but only if you use it to build understanding, not to skip it. The difference between a student who grows and one who does not often comes down to this one choice.

What AI is for

- Understanding what a problem is asking before you start.
- Learning how specific PHP functions work, with isolated examples.
- Checking whether your logic (in pseudocode) is correct.
- Finding the bug in code you wrote yourself.
- Getting feedback on finished code to improve it.

What AI is NOT for

- Generating the solution so you can submit it.
- Replacing the thinking you need to do yourself.
- Producing code you cannot explain line by line.

Academic integrity warning: submitting AI-generated code as your own carries the same consequence as copying from a classmate — loss of marks for both parties. The goal of this assignment is not to produce a working file. It is to build a skill. Those are two completely different outcomes.

The one test that matters

After finishing any question, close the screen and explain your solution out loud — every line, in your own words. If you cannot do this, you do not own the code yet. Go back, ask AI to quiz you, and close the gap before submitting.

Part 2 — 10 Advanced Mixed-Topic Questions

All questions must be solved in one file named `solution.php`. Solutions must be separated by clear comments. Each question combines two or more topics.

Q1. Smart name badge generator

Medium

Topics: [\[Strings\]](#) [\[Loops\]](#) [\[Date\]](#)

Store a list of 5 full names (with random extra spaces) in an array. Loop through them and for each name: trim it, convert it to uppercase, then print only names where the total character count (excluding spaces) is greater than 10. Beside each qualifying name, print the current day of the week.

Requirements:

- Use `trim()` and `strtoupper()` on each name.
- Count characters after removing all spaces (not just trimming).
- Use `date()` to get the day name dynamically.
- Only print names where the no-space character count exceeds 10.

Expected output (example):

```
MOHAMMED AL-HASSAN — Saturday
FATIMA IBRAHIM ZAQOUT — Saturday
```

Q2. Birthday countdown with weekday check

Medium

Topics: [\[Date\]](#) [\[Strings\]](#) [\[Arithmetic\]](#)

Store a birthday as "d-m" in a variable. Extract the day and month manually using only position-based string functions (no `explode()`). Calculate how many days remain until the next birthday, handling the case where the birthday has already passed this year. Then determine what day of the week that birthday will fall on, and state whether it is a weekend or weekday.

Requirements:

- Find the "-" position using `strpos()` only.
- Extract parts using `substr()` only — no `explode()`.
- Handle the case where the birthday is earlier in the same year.
- Use `date("N", mktime(...))` to get the ISO weekday number.

Expected output (example):

```
Next birthday: 15-March-2026
Days remaining: 345
Falls on: Thursday — weekend!
```

Q3. Palindrome multiplication filter

Medium-Hard

Topics: [\[Palindrome\]](#) [\[Loops\]](#) [\[Multiplication\]](#)

For a given number N, display its multiplication table from 10 down to 1. For each result, check if that result is a palindrome number (when converted to a string) without using `strrev()`. Mark palindrome results with a [P] label. Only display rows where the multiplier is odd.

Requirements:

- Loop from 10 down to 1.
- Skip even multipliers.
- Implement palindrome check manually using a loop (no `strrev()`).
- Append [P] to palindrome results in the output.

Expected output (example):

```
11 x 9 = 99 [P]
11 x 7 = 77 [P]
11 x 5 = 55 [P]
```

```
11 x 3 = 33 [P]
11 x 1 = 11 [P]
```

Q4. Email domain extractor with pattern display

Medium-Hard

Topics: [\[Strings\]](#) [\[Pattern\]](#) [\[Loops\]](#)

Store 5 email addresses in variables. For each one, extract the username (part before @) and the domain (part after @) without using any string split functions — only position-based functions. Then display a hollow diamond pattern where the number of rows equals the length of the username. Skip any email where the username length is less than 4.

Requirements:

- Use strpos() to find "@" and substr() to extract both parts.
- No explode(), no split(), no str_split().
- Pattern height = strlen of username after trim.
- Hollow diamond: only print * at the edges of each row.

Expected output (example):

```
  *
 * *
*   *
 * *
  *
```

Q5. Swap and sequence builder

Medium

Topics: [\[Swap\]](#) [\[Even/Odd\]](#) [\[Sequence\]](#)

Store two numbers A and B and swap them without using a third variable. After swapping, use the new values of A and B as the start and end of a range. Print all numbers in that range on one line separated by a dash, but skip multiples of 3, skip every 4th even number, and ensure there is no dash at the start or end.

Requirements:

- Swap using arithmetic only (XOR or addition/subtraction).
- Collect qualifying numbers into an array, then implode with "-".
- Track even number count separately to skip every 4th one.

Expected output (example):

2-4-5-7-8-10-11-13-14-16-17-19-20

Q6. Next-Friday countdown calendar

Medium

Topics: **[Date]** **[Loops]** **[Pattern]**

Print today's date in the format "Today is: Thursday, 14-August-2025". Calculate the days until next Friday (always at least 1, even if today is Friday). Then print a countdown calendar: for each day from today to next Friday, print the day name and date, followed by a separator line of dashes whose length equals the ISO weekday number of that day (Mon=1 ... Sun=7). Mark the current day with [TODAY].

Requirements:

- If today is Friday, next Friday is exactly 7 days away.
- Use date("N") for ISO weekday numbers.
- Loop through each day using mktime() or strtotime().
- Separator length = day's ISO number (e.g., Wednesday = 3 dashes).

Expected output (example):

```
Saturday [TODAY]
-----
Sunday
-----
...
Friday
-----
```

Q7. Full pipeline — name to code to verify

Hard

Topics: **[Strings]** **[Palindrome]** **[Multiplication]** **[Date]**

Given a full name with extra spaces: trim and uppercase it, then remove all vowels (A, E, I, O, U). Count the remaining characters. If that count is even, check if the count value is itself a palindrome number. If the count is odd, print the multiplication table for that count (odd multipliers only, from 10 down to 1). Finally, calculate days until next Friday and append that count to the last output line, separated by a dash — with no trailing dash.

Requirements:

- Vowel removal using a loop or `str_replace` — your choice.
- Palindrome check on the integer count, not a word.
- Odd multiplier table from 10 down to 1.
- Append days-to-Friday to the last line of output.

Expected output (example):

```
Processed name: MHMD ZQLM
Character count: 8 (even)
8 is not a palindrome.
Days until Friday: 3
Result: 8-3
```

Q8. Number grid with diagonal palindrome scan

Hard

Topics: [\[Loops\]](#) [\[Palindrome\]](#) [\[Pattern\]](#)

Generate a 5x5 number grid where cell `[row][col] = row * col`. Print the grid with aligned columns. Then scan the main diagonal (top-left to bottom-right) and collect the diagonal values into a string. Check if that string is a palindrome without using `strrev()`. Finally, print all even numbers from the grid (no duplicates) on one line separated by dashes, skipping every 3rd even number found.

Requirements:

- Grid: rows and cols both go from 1 to 5.
- Use `printf()` or `str_pad()` for column alignment.
- Diagonal values: `[1][1]`, `[2][2]`, `[3][3]`, `[4][4]`, `[5][5]`.
- Deduplicate even numbers before applying the skip rule.

Expected output (example):

```
1  2  3  4  5
2  4  6  8 10
...
Diagonal string: 14925 — not a palindrome
Even numbers: 2-4-8-10-16-20-24-25
```

Q9. Email report with birthday cross-check

Hard

Topics: [\[Strings\]](#) [\[Date\]](#) [\[Loops\]](#)

Store 4 email addresses and 4 corresponding birthdays ("d-m") in parallel arrays. Loop through them together. For each entry: extract the username from the email using only substr() and strpos(), check if the username is a palindrome (no strrev()), calculate days until that birthday, and print a formatted report line. At the end, print which username has the soonest upcoming birthday.

Requirements:

- Parallel arrays — same index = same person.
- Username palindrome check: implement manually.
- Birthday days calculation must handle already-passed dates.
- Find and print the minimum day's value at the end.

Expected output (example):

```
madam@test.com | Username: madam [PALINDROME] | Birthday in 12 days
level@iug.edu  | Username: level [PALINDROME] | Birthday in 88 days
---
Soonest birthday: madam (12 days)
```

Q10. Grand finale — the complete PHP dashboard

Very Hard

Topics: [\[All Topics\]](#) [\[Logic\]](#) [\[Output\]](#)

Build a single PHP script that outputs a formatted text "dashboard" combining everything. It must: (1) Print a header with today's full date and days until next Friday. (2) Accept a name, process it (trim, uppercase, remove vowels, count chars, palindrome check on count). (3) Generate the multiplication table for a number N (odd multipliers only, 10 down to 1) and mark any result that is a palindrome. (4) Print even numbers 0–50 skipping every 3rd, on one line with dashes. (5) Extract username from a given email. (6) Swap two numbers and print the swapped result. (7) Print a hollow diamond of height equal to the vowel-free character count, max 9. (8) Print numbers 1–30 skipping multiples of 4, on one line with dashes.

Requirements:

- Each section must be separated by a visible comment/divider in output.
- No function definitions — solve everything inline.
- All palindrome checks must be done without strrev().
- All string splits must use only position-based functions (no explode).

Expected output (example):

```
=== PHP DASHBOARD ===  
Date: Saturday, 04-April-2026 | Days to Friday: 6  
...  
[Section 2] Name: MHMD | Count: 4 | Palindrome: NO  
...
```

Part 3 — AI Prompt Guide: Step by Step

This section gives you the exact type of prompt to use at each stage of solving a problem. Replace the bracketed placeholders with your actual content. The prompts are ordered to mirror a healthy problem-solving workflow.

Step 1 — Understanding the problem before writing any code

Read the question carefully twice. Identify: what is the input, what is the output, and what transformations happen in between. Only then ask AI for conceptual help.

Prompts to use:

```
Explain to me in plain English what this PHP question is asking me to do:  
[paste the question].
```

```
What PHP concepts and functions would I need to know to solve this problem?  
Do not write the code yet.
```

```
Can you break this problem down into a numbered list of smaller steps?
```

Step 2 — Learning the specific PHP functions you need

Before writing code, understand the tools. Ask AI to explain specific PHP functions you'll need, with small isolated examples.

Prompts to use:

```
How does PHP's strpos() function work? Show me a simple example with a  
string and a character to find.
```


What is the difference between `trim()` and `str_replace()` when removing spaces in PHP?

How do I get the day of the week in PHP using the `date()` function? What format characters should I use?

How does PHP calculate days between two dates? Show me the `mktime()` approach.

Step 3 — Planning your solution (pseudocode first)

Write your solution in plain English or pseudocode before touching PHP. Then ask AI to verify your logic — not write your code.

Prompts to use:

Here is my plan to solve the problem in pseudocode: [paste your plan]. Is my logic correct? What am I missing?

I plan to check palindromes like this: reverse the string manually using a loop and compare. Will that approach work in PHP without `strrev()`?

My plan for swapping two variables without a third: $A = A + B$, $B = A - B$, $A = A - B$. Is this correct?

Step 4 — Debugging your own code

Write your attempt first, even if it is broken. Then share it with AI and describe the specific problem you see.

Prompts to use:

Here is my PHP code: [paste]. It is supposed to print a hollow diamond but instead prints a filled one. What is wrong with my loop condition?

My multiplication table prints even multipliers even though I wrote a condition to skip them. Here is my code: [paste]. What did I get wrong?

My output has an extra dash at the end of the line. Here is how I am building the string: [paste]. How do I fix this without removing the logic?

I am getting a PHP error: [paste error]. What does this error mean and where in my code should I look?

Step 5 — Reviewing and improving your finished solution

After your code works, ask AI to review it. Treat this like getting feedback from a senior developer.

Prompts to use:

My code works but I want to know if there is a cleaner or more efficient way. Here it is: [paste]. Do not rewrite it — just tell me what you would improve and why.

Are there any edge cases my solution might miss? The problem says: [paste question]. My code: [paste].

Can you check my comments and suggest better ones? I want them to explain the WHY, not just the WHAT.

What PHP best practices am I not following in this code? [paste]

Step 6 — Testing your understanding (self-check)

The final test: can you explain your own code? Use AI to quiz you — this is the most honest check of whether you have truly learned.

Prompts to use:

Ask me questions about this PHP code to check if I really understand it: [paste your code]. Quiz me one question at a time and wait for my answer.

I will explain this code line by line and you tell me if I am correct or if I am missing something: [paste code and your explanation].

What would happen to my code if [describe an edge case — e.g., the name is empty, or the number is 0]? Let me guess first: [your guess]. Am I right?

Final reminder: AI is a mirror, not a mind. It reflects back whatever thinking you bring to it. Bring good thinking — careful reading, honest attempts, specific questions — and AI will sharpen your skills. Bring none of those, and AI will only produce the illusion of learning.

You can use any AI model from your choice

The deadline for submission is Sunday, 12/4/2026

Good luck 🙌